RAP: Refine a Prediction of Protein Secondary Structure

Ying-Chuan Liu[1], Chun-Yuan Lin[2], Kun-Ming Yu[3] and Chuan-Yi Tang[1*]
[1]Department of Computer Science and [2]Institute of Molecular and Cellular Biology, National Tsing Hua University,
[3]Department of Computer Science and Information Engineering, Chung Hua University, Hsinchu, Taiwan 300, ROC
{g934313@oz, cyulin@mx, *cytang@cs}.nthu.edu.tw, yu@chu.edu.tw

Abstract

RAP aims to refine protein secondary structure prediction from one of famous prediction tools. Protein secondary structure prediction has been extensively discussed for almost 50 years and the machine learning is one of feasible methods for it with more than 70% accuracy. PSIPRED, PHD and PROF are well-known machine learning approaches and based on the three-state prediction: helix, strand, and coil. RAP is a post-processing tool with logical and statistic methods to refine the scores of three-state prediction. Hence, RAP can be easily applied to any protein secondary structure prediction tool if it uses three-state prediction. RAP was tested on the CASP data set with 181 targets and a large-scale data set with 69534 chains of 31402 proteins in PDB database. In the experiment, PSIPRED is used to give the scores of three-state prediction for each test data at first and then RAP predicts the result by refining the scores from PSIPRED. RAP obtains overall prediction accuracy rates 81.3% and 80.7 % for the CASP and PDB data sets, respectively.

1 Introduction

Large-scale sequencing projects have produced a massive number of proteins with putative amino acid sequences. In contrast, much fewer are known in terms of protein tertiary structures. Some famous structure databases, e.g., SCOP [1] and CATH [2], contribute only no more than 33000 entries in PDB [3] database (SCOP release version 1.69: 25973 PDB entries in Jul. 2005, PDB: 32355 entries in 23-Aug. 2005). This is only about 16% of collections in the Swiss-Port (release version 49.0: 207132 entries in 07-Feb-2006). Physically, x-ray diffraction or NMR is used to determine the tertiary structure for a protein. However, each has its limitation [4]. Therefore, extracting structural information from sequence databases becomes an important and complementary alternative, especially for swiftly determining protein functions or discovering new compounds for medical or therapeutic purposes. An accurate method for protein secondary structure prediction could facilitate the prediction of tertiary structure [5, 6].

Protein secondary structure prediction has been studied for almost 50 years [7]. Many methods, e.g., stereochemical principles [8], statistics [9] and multiple sequence information [10], have been proposed for protein secondary structure prediction. The machine learning, e.g., neural networks and support vector machines [11], is one of feasible prediction methods for protein secondary structure with more than 70% accuracy [12]. PHD [12] is a well-known machine learning approach using neural networks and based on the three-state prediction, helix (H), strand (E) and coil (C). For a protein, PHD assigns a value as a probability to each state at first and then the prediction result is determined by one of states with the largest value. This way is followed by most of later prediction tools based on the machine learning approach. For example, PSIPRED [13] and PROF [7] which are under the three-state prediction are both more popular than PHD and adopt more information, new observations and machine learning techniques. They have achieved higher accuracy (almost 80% accuracy) than PHD for predicting protein secondary structure. Although more machine learning approaches [14-18] have been proposed in recent years, their prediction accuracy is close to PSIPRED or PROF. This shows that it may be hard to design a new machine learning system which has better accuracy than existing prediction tools. Therefore, refining prediction results from a famous tool is more beneficial than designing a new system.

In this paper, we present a method, RAP, of protein secondary structure prediction by refining the prediction results from a famous tool. For an objective protein, RAP uses the scores of three-state prediction from a tool as an input at first and then adopts a logical method and a statistic method to refine the input. Finally, RAP predicts the protein secondary structure for the objective protein by refined data. Hence, RAP is a simple tool comparing to a new system design and can be applied to various tools, e.g., PHD, PROF and PSIPRED. In order to evaluate RAP, a CASP data set with 181 targets and a large scale data set with 69534 chains of 31402 proteins in PDB database are tested in RAP.

* The corresponding author.

PSIPRED is used to obtain scores of three-state prediction for each protein in the experiment.

2 Materials and methods

The major components of RAP for protein secondary structure prediction include: (1) combining scores of three-state prediction, called RAP_S, (2) normalizing the combining scores from RAP_S by rank, called RAP_R, and (3) voting for the predictions results by RAP_S and RAP_R, called RAP_V. Each of RAP_S, RAP_R and RAP_V can predict the protein secondary structure for an objective protein individually.

2.1 RAP_S: combining scores of three-state prediction

For a tool with three-state prediction, e.g., PSIPRED, the secondary structure for a residue of an objective protein is determined by one of states with the largest score. For each state, this tool assigns a score as a probability to it according to its method. Each score is scaled onto 0-9. In the logic, it may be unsuitable for doing a decision, which is determined by an umpire with his comment, with more than two umpires. For example, a paper should not be accepted if it has a weakly accepted comment and two strongly rejected comments from three referees. RAP_S is used to combine scores of three-state prediction for each state. Assume that a protein sequence $S = s_1s_2\ldots s_n$ with a length of n and its corresponding scores of three-state prediction is a $3\times n$ matrix M, where m_{ij} in M is a score of the residue s_j in the class i, $i\in\{H,E,C\}$ and $1\le j\le n$. The combination process is

$$t_{p,q} = \begin{cases} m_{p,j} & \text{if } p=q \\ 9-m_{p,j} & \text{otherwise.} \end{cases}$$

$$score_{p,j} = \frac{\sum_{q=1}^{3} t_{p,q}}{3}$$

where, $p,q\in\{H,E,C\}$ and $0\le m_{p,q}\le 9$. T is a 3×3 temporary matrix, and *SCORE* is a $3\times n$ matrix with combining scores. For an objective protein, RAP_S can predict the secondary structure by the combining scores.

2.2 RAP_R: normalizing the combining scores from RAP_S by rank

After RAP_S, the combining scores of three-state prediction for a residue may still be unreliable. The reason is that the strength of the original score in one of states may be amplified. For example, there is a student takes two courses. Assume that a teacher in one of courses always gives his students the grade A and another teacher in the other course only gives the best student with the grade B. If this student gives the grade B from the later teacher and the grade A from the former teacher, it can not indicate that this student take more effort in the former course than that in the later course. This effect can not be eliminated by RAP_S. In RAP_R, it computes the occurrence of each revised score in each state at first. Then RAP_R normalizes the distribution of the occurrence of each revised score in each state. Let $L = l_1l_2\ldots l_k$ with a length of k be a sorted list of all revised scores that ever appear in matrix *SCORE* and $x_i = (x_{i,1}, \ldots, x_{i,k})$ be the occurrence of l_v in $score_i$ where $1\le v\le k$ and $score_i$ is the row vector of *SCORE* for class i. Afterwards, the quantile normalization [19] is exploited to get x'_i by normalizing the distribution of x_i. Then, all values in x'_i are transformed to ranks as follows:

$$x''_{i,v} = \begin{cases} 1 & \text{if } v=k \\ x''_{i,v+1} + x'_{i,v+1} & \text{otherwise.} \end{cases}$$

$$rank_{p,j} = f(score_{p,j})$$

where *RANK* is a 3×n matrix that stores the rank of each value in *SCORE* and $f(x_{p,q})$ is a function that find v which is the position of $x_{p,q}$ in L and return $x''_{p,v}$ which is the rank of $score_{p,j}$. For an objective protein, RAP_R can predict the secondary structure by the ranks.

2.3 RAP_V: voting for the predictions results by RAP_S and RAP_R

RAP_V is used to combine prediction results of RAP_S and RAP_R with *SCORE* and *RANK*, which are under different measurement. Before the combination, $score_{p,q}$ ranks as $rs_{p,q}$ among the qth column vector of *SCORE*, and $rank_{p,q}$ ranks as $rr_{p,q}$ among the qth column vector of *RANK*, where *RS* and *RR* are two 3×n matrix. A jury decision is used to combine RAP_S and RAP_R and is formulated as follows:

$$vote_{p,j} = \frac{rs_{p,j} + rr_{p,j}}{2}.$$

For an objective protein, RAP_V can predict the secondary structure by the voting results. For RAP_S, RAP_R and RAP_V, if at least two revised scores or ranks or voting results are equal, a selection priority, C>H>E provided by Wu *et al.* [15], is adopted here. It might be improper to use the priority for all proteins, e.g. all-alpha proteins. Another intuitive selection priority H>E>C is also applied into RAP.

Table 1. The prediction results of PSIPRED in four classes are shown. The value in parentheses represents the number of protein in that class.

	CASP					PDB				
	all-α (24)	all-β (4)	α-β (46)	Other (107)	All (181)	all-α (7318)	all-β (2886)	α-β (12163)	Other (47167)	All (69534)
Q_α	91.93	22.23	85.05	81.11	82.24	91.77	40.13	85.42	74.56	76.84
Q_β	19.83	77.25	77.5.	75.45	68.64	23.19	77.15	76.11	69.83	66.32
Q_c	74.37	80.15	79.93	77.12	77.54	76.61	82.09	79.15	80.71	80.06
Q_3	86.25	76.35	81.41	80.31	81.29	86.83	78.39	80.77	79.69	80.58

Table 2. The prediction results of RAP in four classes are shown. x^S: x is the prediction accuracy of RAP_S; x^R: x is the prediction accuracy of RAP_R; x^V: x is the prediction accuracy of RAP_V.

	CASP					PDB				
	all-α (24)	all-β (4)	α-β (46)	Other (107)	All (181)	all-α (7318)	all-β (2886)	α-β (12163)	Other (47167)	All (69534)
Q_α	92.66^S	25.00^R	88.45^V	85.02^V	85.58	92.28^S	68.68^R	87.75^V	79.94^V	82.14
Q_β	41.70^R	75.55^S	83.43^R	81.39^R	76.52	51.30^R	75.26^S	82.32^R	77.87^R	75.74
Q_c	90.54^V	83.03^V	82.64^S	79.76^S	81.99	92.54^V	86.52^V	81.65^S	83.22^S	84.06
Q_3	86.36^S	76.10^S	81.35^S	80.34^S	81.30	86.93^S	77.95^S	80.74^S	79.82^S	80.65

2.4 Data sets

Two data sets are used to evaluate RAP. The first data set was extracted from CASP2 to CASP6 experiments [20-24] with 181 targets that were marked with PDB IDs and not canceled by CASP. The second one was obtained from PDB database in which there were 31402 proteins with 69534 chains. We collected these data to run PSIPRED, and then took scores of three-state prediction to execute the RAP. The reference secondary structure states for each structure were derived from the definitions produced by DSSP [25]. The eight states (H, I, G, E, B, S, T, _) were reduced to three states according to the rules outlined by PSIPRED [13].

3 Experiments

Running PSIPRED is the most time-consuming process in the whole prediction and its execution time increases with the length of a protein sequence. According to the experiment results, the execution time of PSIPRED for the PDB data set is over 130 CPU days on a Linux system of AMD Athlon PC with a clock rate of 2G MHz and 1GB memory. Thus, the PDB data set was distributed and predicted by PSIPRED among a 17-CPU PC-cluster in order to decrease the time requirement. After running PSIPRED, RAP is a fast tool, which takes less than 20 minutes, to refine the scores of three-state prediction for the PDB data set.

The standard percentage accuracy rate Q_i is used to evaluate RAP, which is given by

$$Q_i = \frac{p_i}{l_i} \times 100\%$$

where l_i is the number of testing data in the class i, $i \in \{H(\alpha), E(\beta), C(c)\}$ and p_i is the number of data being correctly predicted in the class i. The overall prediction accuracy rate Q_3 is given by

$$Q_3 = \sum_{i=1}^{n} q_i Q_i$$

where $q_i = l_i/K$, where K is the total number of testing data, and $n = 3$. To identify the prediction ability for different proteins, each protein was classified into one of the following structural classes: all-alpha, all-beta, alpha-beta and other [7]. As shown in Table 1, for the CASP and PDB data sets, PSIRPED performs well on Q_α for the all-alpha class and on Q_β for the all-beta class; nevertheless, Q_β for the all-alpha class and Q_α for the all-beta class are far below average for all proteins. It implies that ignoring the composition of secondary structure might lead to the reduction of accuracy.

Table 2 summarizes 16 rules according to the experiment results of testing RAP_S, RAP_R and RAP_V on the four classes of the CASP data set, respectively. For instance, if biologists would like to gives as many α-helices as possible for a protein sequence belongs to the all-alpha class, RAP_S is the most suitable among the three approaches. Note that RAP made a 7.88% improvement over PSIPRED on Q_β, and upgraded Q_α and Q_c to 85.58% and 81.99%, respectively. By exploiting the same rules on the PDB data set, Q_3 was increased from 80.58% (PSIPRED) to 80.65% (RAP). Also, Q_β of RAP achieved 75.74% which was an improvement of 9.42% over PSIPRED. Figure 1 shows the distributions of Q_α, Q_β, Q_c and Q_3 from RAP for the PDB data set. There are near 90% of the PDB data set whose Q_3 scores reach better than 70% ; nonetheless, Q_α for 8% of the proteins and Q_β for 15% of the proteins are less than 10%.

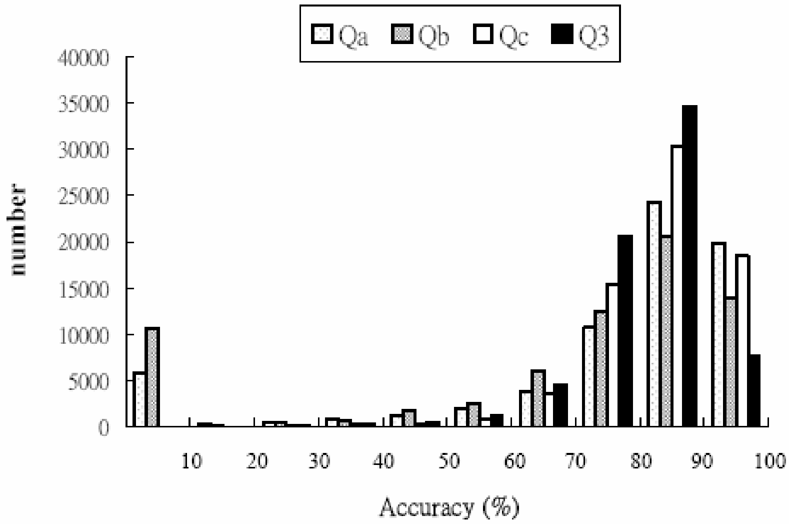

Fig 1. Bar graph showing the distribution of Q_α, Q_β, Q_c and Q_3 for the PDB data set.

4 Discussion

According to the experimental results, it is clear that an accurate classification holds the key to make improvements. Previous studies in the structural classification field have achieved better than 70% accuracy from the amino acid composition [26, 27], i.e. the predictions of less than five classes [28]. In general, PSIPRED is an accurate method of protein secondary structure prediction. For the all-alpha class, however, Q_β of PSIPRED is much lower than Q_α; similarly, Q_α is fairly low over the all-beta class. In those cases, RAP_R is able to greatly enhance Q_β for the all-alpha class and Q_α for the all-beta class. Furthermore, Q_α and Q_β for the alpha-beta class are upgraded by RAP_V and RAP_R, respectively. On the other hand, there is still room for improvement on Q_α and $Q_\beta$of RAP, i.e. for those whose accuracy is below 10%.

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